Adnan Ahmad

AI / Machine Learning Engineer • Computer Vision • NLP / LLMs

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SUMMARY

Machine Learning Engineer with a PhD in ML and 13+ peer-reviewed publications, specializing in building and deploying AI systems across **computer vision, NLP/LLMs, and multimodal deep learning**. Hands-on experience shipping real-time inference pipelines (PyTorch, FastAPI, Docker, GCP), retrieval-augmented generation (RAG) systems, and large-scale distributed training. Seeking a Computer Vision / NLP / AI Engineer role turning research-grade models into reliable products.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java
ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, CNNs, Transformers, Representation / Continual / Self-Supervised Learning, Reinforcement Learning
NLP & LLMs: Hugging Face Transformers, RAG, LangChain, SentenceTransformers, LLM Fine-Tuning (LoRA / PEFT), Prompt Engineering, Vector Databases (Qdrant)
Computer Vision: OpenCV, Image Classification & Detection, Multimodal (image / signal) Modeling, Medical Image Analysis
MLOps & Cloud: Docker, FastAPI, Git, MLflow, Weights & Biases, CUDA, Google Cloud Platform (Vertex AI, Compute Engine)
Data: Pandas, NumPy, Data Pipelines, Distributed / Federated Training, Real-Time Streaming (LSL)

EXPERIENCE

AI / Machine Learning Scientist — March 2024 – May 2026

Deakin University — Applied ML & Human-AI Systems Lab — *Waurn Ponds, VIC, Australia*

- Built and deployed **real-time multimodal ML systems** that process live physiological signal streams (EEG/ECG/eye-tracking) through low-latency inference pipelines, served as containerized microservices on GCP (Docker, FastAPI).
- Developed **continual-learning** models for non-stationary data streams, improving cross-subject generalization and reducing accuracy degradation under distribution shift.
- Designed **reinforcement-learning** agents for human-AI collaboration using hierarchical skill learning, improving task coordination in both simulation and human-in-the-loop deployments; results published in 4+ papers.

Machine Learning Scientist (Federated & Distributed Learning) — July 2024 – Jan 2025

National Research Council of Italy (IIT-CNR) — ERCIM Fellowship — *Pisa, Italy*

- Designed **decentralized federated-learning** algorithms for peer-to-peer networks handling data, model, and communication heterogeneity with no central coordinator.
- Built a second-order (Hessian-based) aggregation method in PyTorch that **reduced communication rounds and sped up convergence** versus FedAvg on benchmark vision datasets; work under review at IEEE TETCI.

Machine Learning Teaching Assistant — March 2021 – March 2024

Deakin University — *Burwood, VIC, Australia*

- Led workshops on deep learning, **computer vision, and NLP**, plus core AI (search, optimization, agent-based reasoning) for cohorts of postgraduate students.

PROJECTS

- **InsightDocs — Production Agentic RAG** — Deployed retrieval-augmented generation system with **hybrid retrieval (dense + BM25) and cross-encoder reranking**, an agentic router and query decomposition, and a quantitative evaluation harness; serves grounded, cited answers via FastAPI + Qdrant + Docker. GitHub
- **MedVision — Explainable Skin-Lesion Classifier** — Fine-tuned a ConvNeXt classifier on DermaMNIST (7 classes) to **0.91 macro-AUROC** with **Grad-CAM** explanations; optimized inference via ONNX + INT8 (**3.4× faster, 3.9× smaller**), served through a Dockerized FastAPI app. GitHub
- **Real-Time Human–AI Coordination via Physiological Feedback** — Real-time system where models infer human mental state from EEG/ECG/eye-tracking and RL agents adapt in response; built on LSL streaming, Docker, and FastAPI. Project Page | GitHub

EDUCATION

PhD in Machine Learning — Deakin University, Australia — *Dec 2020 – Sep 2024*
Thesis: Robust and Communication-Efficient Federated Learning under Non-IID Data.
BSc in Electrical (Computer) Engineering — COMSATS University, Pakistan — *Sep 2013 – Sep 2017*

PUBLICATIONS & AWARDS

13+ peer-reviewed papers in Machine Learning, Pattern Recognition, Neurocomputing, ACM TIST, and PAKDD. Reviewer for NeurIPS, ICLR, and Pattern Recognition. Full list: Google Scholar.

Awards: ERCIM "Alain Bensoussan" Fellowship (2024); Chartered Professional Engineer (Engineers Australia); Deakin Cyber Research Grant (AUD 12K); Postgraduate Research Scholarship (2020–2024).